

**Time:** 5 minutes**Outcome:** Distinguish nominal and real variables and analyze long-run neutrality.**Difficulty:**

Intermediate

Monetary Neutrality



THE CORE IDEA

Monetary neutrality means that a lasting change in the money supply changes nominal variables but not real output or employment in the long run. Nominal variables changed, but real output did not. Monetary neutrality means nominal variables can change without changing real variables in the long run. Separate nominal from real variables. A proportional change in nominal wages and prices can leave purchasing power unchanged.



HOW TO RECOGNIZE IT

- Separate nominal variables, measured in money, from real variables, measured in goods, services, or purchasing power.
- A lasting money-supply change can alter the price level and other nominal values in the long run.
- Under monetary neutrality, long-run real output and employment remain determined by real resources and technology.
- Do not apply long-run neutrality to the short run, when sticky prices can make money affect real activity.

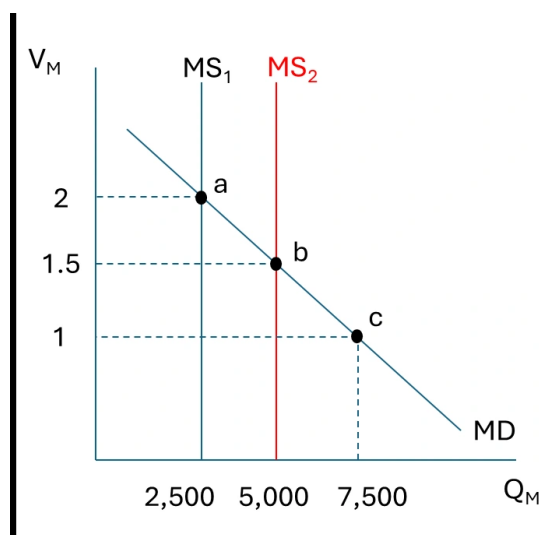


WATCH OUT

Do not assume money changes real variables permanently in the long run. Monetary neutrality means money affects nominal variables in the long run, not real output or employment.



WORKED EXAMPLE: NOMINAL VERSUS REAL EFFECTS



The graph shows money supply rising from 2,500 to 5,000 while money's value falls from 2 to 1.5. The corresponding price-level index rises from $1/2 = 0.50$ to $1/1.5 = 0.67$. In the long-run neutrality model, this nominal adjustment does not by itself raise real output.



CHECK YOURSELF

A student says, "If money is neutral, monetary policy can never affect real output." What is the best correction?

**READY?**

Return to the game and master this concept.